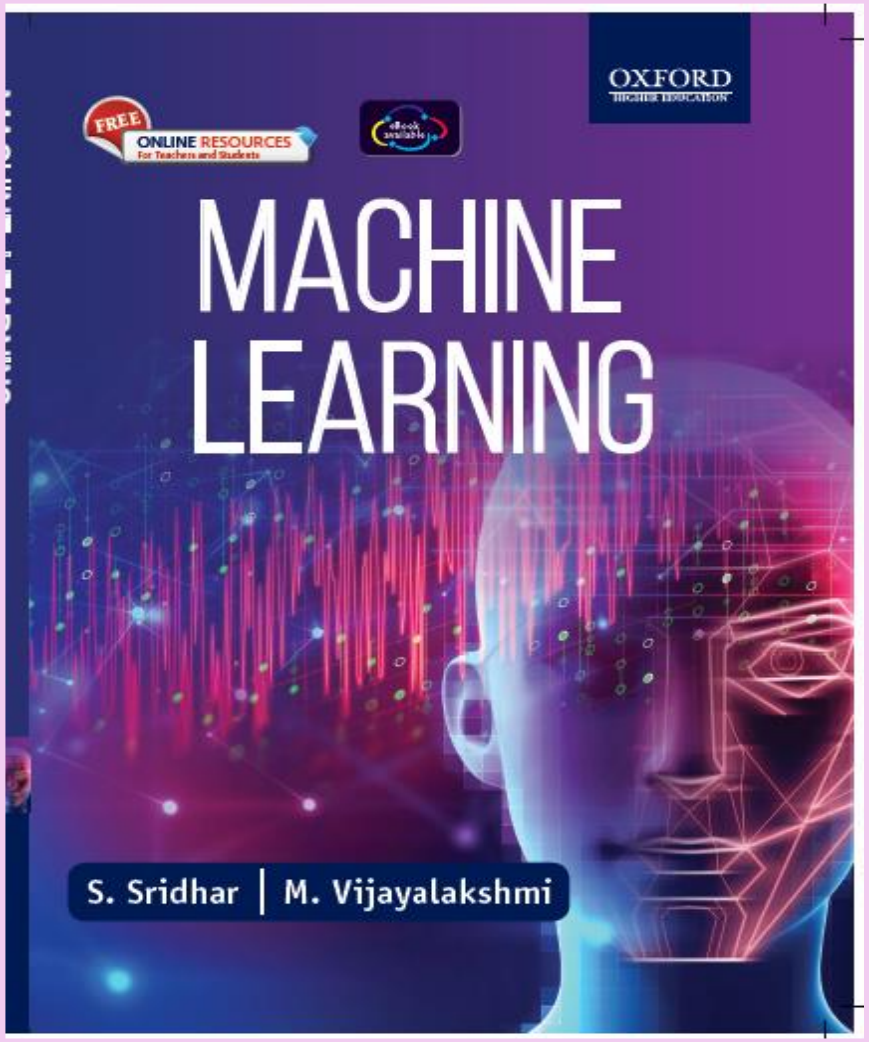


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Machine Learning

S. Sridhar and M. Vijayalakshmi

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Chapter 2

Regression Analysis

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What is Regression Analysis?

- A KIND OF SUPERVISORY LEARNING
- Y=F(X)

HERE, Y IS CALLED DEPENDENT VARIABLE AND X IS CALLED INDEPENDENT VARIABLE.

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Focus of Regression Analysis

1. What is the relationship between the variables?
2. What is the strength of the relationships?
3. What is the nature of the relationship such as linear or non-linear?
4. What is the relevance of the attributes?
5. What is the contribution of each attribute?

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Applications of Regression Analysis

There are many applications of regression analysis. Some of the applications of regressions include predicting:

1. Sales of a goods or services
2. Value of bonds in portfolio management
3. Premium on insurance companies
4. Yield of crops in agriculture
5. Prices of real estate

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Correlation

CORRELATION BETWEEN TWO VARIABLES CAN BE FOUND USING A SCATTER PLOT

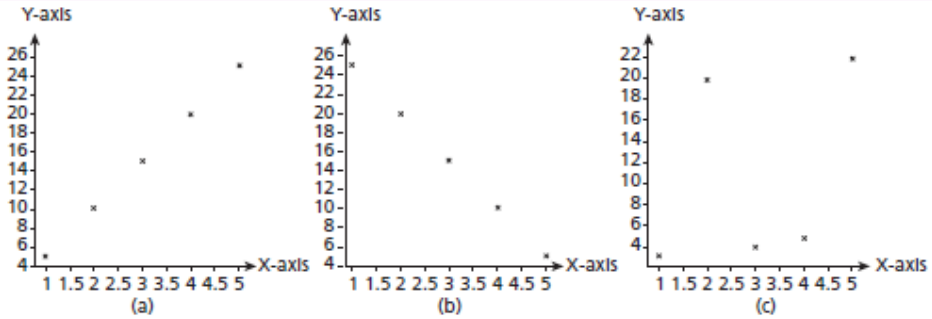


Figure 5.1: Examples of (a) Positive Correlation (b) Negative Correlation (c) Random Points with No Correlation

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Causation

1. CAUSATION IS ABOUT RELATIONSHIP BETWEEN TWO VARIABLES AS X CAUSES Y. THIS IS CALLED X IMPLIES B.

2. REGRESSION IS DIFFERENT FROM CAUSATION.

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Linear and Non-Linear Relationships

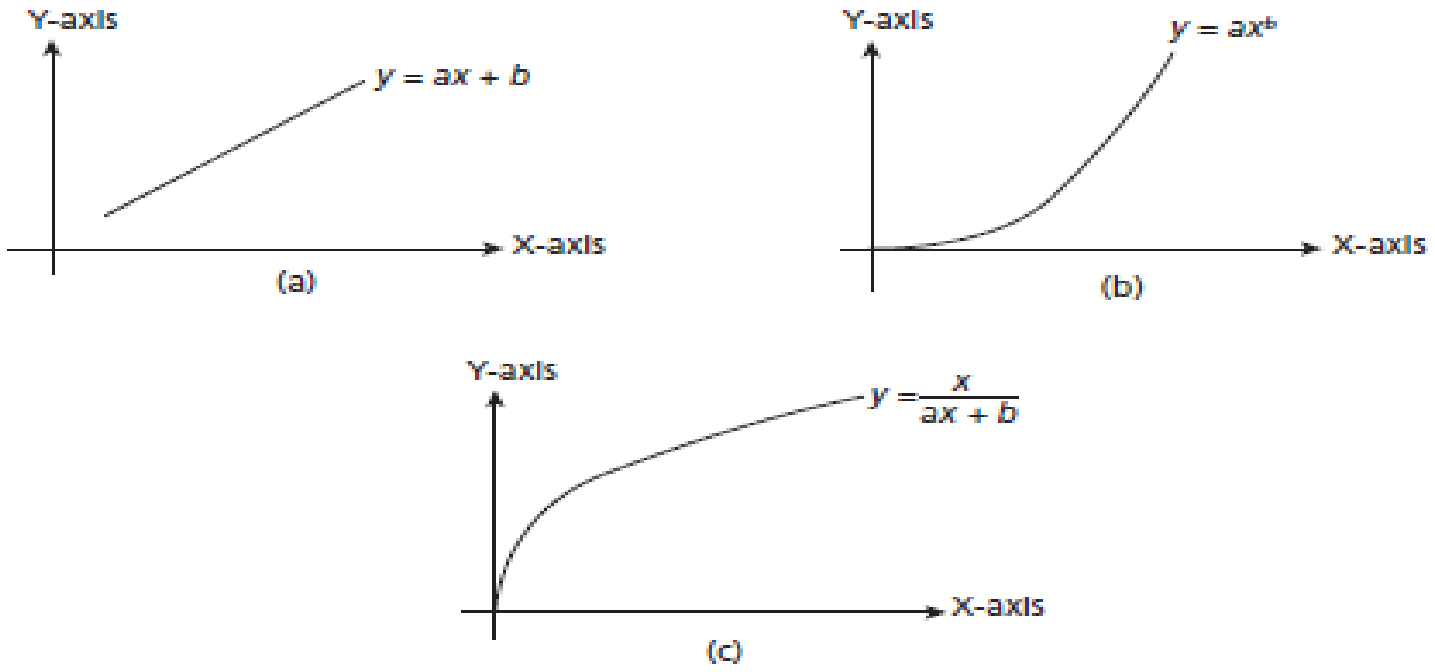


Figure 5.2: (a) Example of Linear Relationship of the Form $y = ax + b$ (b) Example of a Non-linear Relationship of the Form $y = ax^b$ (c) Examples of a Non-linear Relationship $y = \frac{x}{ax + b}$

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Types of Regression

Types of Regression Methods

The classification of regression methods is shown in Figure 5.3.

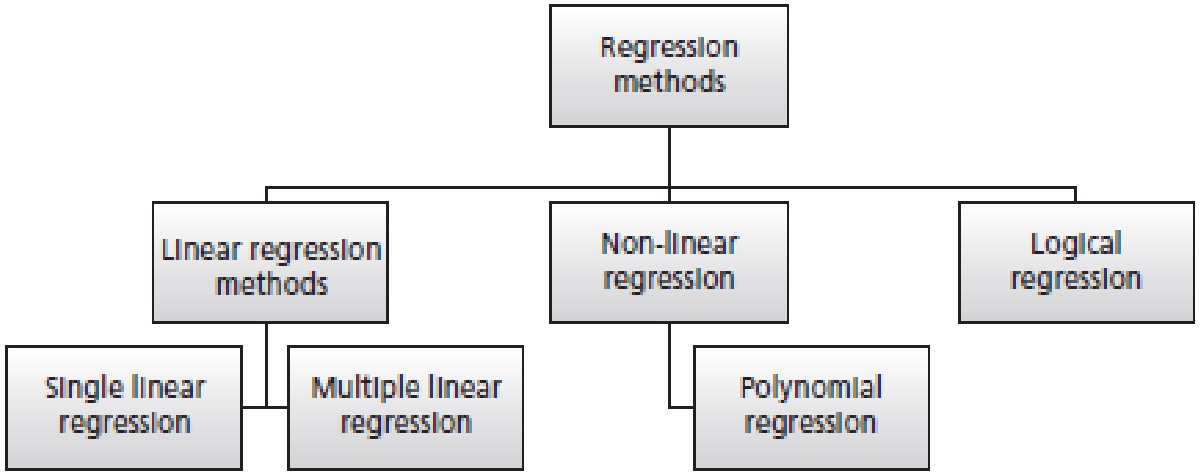


Figure 5.3: Types of Regression Methods

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Limitations of Regression

1. Outliers – Outliers are abnormal data. It can bias the outcome of the regression model, as outliers push the regression line towards it.
2. Number of cases – The ratio of independent and dependent variables should be at least 20 : 1. For every explanatory variable, there should be at least 20 samples. Atleast five samples are required in extreme cases.
3. Missing data – Missing data in training data can make the model unfit for the sampled data.
4. Multicollinearity – If exploratory variables are highly correlated (0.9 and above), the regression is vulnerable to bias. Singularity leads to perfect correlation of 1. The remedy is to remove exploratory variables that exhibit correlation more than 1. If there is a tie, then the tolerance (1 – *R* squared) is used to eliminate variables that have the greatest value.

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Linear Regression

$$y = a_0 + a_1 \times x + e$$

The assumptions of linear regression are listed as follows:

1. The observations (y) are random and are mutually independent.
2. The difference between the predicted and true values is called an error. The error is also mutually independent with the same distributions such as normal distribution with zero mean and constant variables.
3. The distribution of the error term is independent of the joint distribution of explanatory variables.
4. The unknown parameters of the regression models are constants.

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Ordinary Least Square Approach

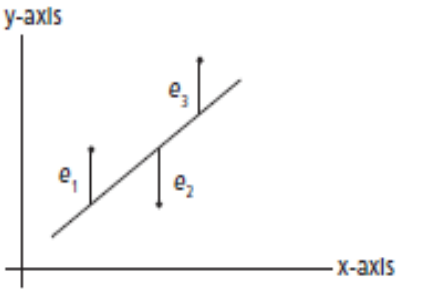


Figure 5.4: Data Points and their Errors

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Linear Regression Model

$$J(a_1, a_0) = \sum_{i=1}^{n}[y_i - f(x_i)]^2$$
$$= \sum_{i=1}^{n}[y_i - (a_0 + a_1 x_i)]^2$$

$$a_1 = \frac{(\overline{xy}) - (\overline{x})(\overline{y})}{(\overline{x_i^2}) - (\overline{x})^2} \quad (5.9)$$

And the value of a_0 is given as follows:

$$a_0 = (\overline{y}) - a_1 \times \overline{x} \quad (5.10)$$

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Linear Regression Example

Example 5.1: Let us consider an example where the five weeks' sales data (in Thousands) is given as shown below in Table 5.1. Apply linear regression technique to predict the 7th and 9th month sales.

Table 5.1: Sample Data

x_i (Week)	y_i (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

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Linear Regression Example

Table 5.2: Computation Table

x_i	y_i	$(x_i)^2$	$x_i \times y_i$
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19
Sum = 15 Average of (x_i) $= \overline{x} = \frac{15}{5}$ $= 3$	Sum = 12.6 Average of (y_i) $= \overline{y} = \frac{12.6}{5}$ $= 2.52$	Sum = 55 Average of (x_i^2) $= \overline{x_i^2} = \frac{55}{5}$ $= 11$	Sum = 44.4 Average of $(x_i \times y_i)$ $= \overline{xy} = \frac{44.4}{5}$ $= 8.88$

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Linear Regression Example

Let us compute the slope and intercept now using Eq. (5.9) as:

$$a_1 = \frac{8.88 - 3(2.52)}{11 - 3^2} = 0.66$$

$$a_0 = 2.52 - 0.66 \times 3 = 0.54$$

The fitted line is shown in Figure 5.5.

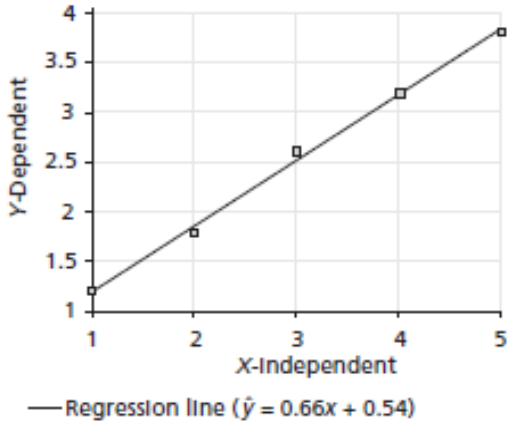


Figure 5.5: Linear Regression Model Constructed

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Linear Regression Example

Let us model the relationship as $y = a_0 + a_1 \times x$. Therefore, the fitted line for the above data is: $y = 0.54 + 0.66 \times x$.

The predicted 7th week sale would be (when $x = 7$), $y = 0.54 + 0.66 \times 7 = 5.16$ and the 12th month, $y = 0.54 + 0.66 \times 12 = 8.46$. All sales are in thousands.

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Linear Regression in Matrix Form

Linear Regression in Matrix Form

Matrix notations can be used for representing the values of independent and dependent variables. This is illustrated through Example 5.2.

The Eq. (5.3) can be written in the form of matrix as follows:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} \tag{5.11}$$

This can be written as:

$Y = Xa + e$, where X is an $n \times 2$ matrix, Y is an $n \times 1$ vector, a is a 2×1 column vector and e is an $n \times 1$ column vector.

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Linear Regression in Matrix Form

Example 5.2: Find linear regression of the data of week and product sales (in Thousands) given in Table 5.3. Use linear regression in matrix form.

Table 5.3: Sample Data for Regression

x_i (Week)	y_i (Product Sales in Thousands)
1	1
2	3
3	4
4	8

Solution: Here, the dependent variable X is be given as:

$$x^T = [1\ 2\ 3\ 4]$$

And the independent variable is given as follows:

$$y^T = [1\ 3\ 4\ 8]$$

The data can be given in matrix form as follows:

$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$. The first column can be used for setting bias.

and $Y = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 8 \end{pmatrix}$

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Linear Regression in Matrix Form

The regression is given as:

$$a = ((X^T X)^{-1} X^T) Y$$

The computation order of this equation is shown step by step as:

1. Computation of $(X^T X) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \times \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}$

2. Computation of matrix inverse of $(X^T X)^{-1} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}^{-1} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{pmatrix}$

3. Computation of $((X^T X)^{-1} X^T) = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0.5 & 0 & -0.5 \\ -0.3 & -0.1 & 0.1 & 0.3 \end{pmatrix}$

4. Finally, $((X^T X)^{-1} X^T) Y = \begin{pmatrix} 1 & 0.5 & 0 & -0.5 \\ -0.3 & -0.1 & 0.1 & 0.3 \end{pmatrix} \times \begin{pmatrix} 1 \\ 3 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} -1.5 \\ 2.2 \end{pmatrix} \begin{pmatrix} Intercept \\ slope \end{pmatrix}$

Thus, the substitution of values in Eq. (5.11) using the previous steps yields the fitted line as $2.2\,x - 1.5$.

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Validation of Regression Methods

Standard Error

Residuals or error is the difference between the actual (y) and predicted value ($\hat{y}$).

If the residuals have normal distribution, then the mean is zero and hence it is desirable. This is a measure of variability in finding the coefficients. It is preferable that the error be less than the coefficient estimate. The standard deviation of residuals is called residual standard error. If it is zero, then it means that the model fits the data correctly.

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Validation of Regression Methods

Mean Absolute Error (MAE)

MAE is the mean of residuals. It is the difference between estimated or predicted target value and actual target incomes. It can be mathematically defined as follows:

$$\text{MAE} = \frac{1}{n}\sum_{i=0}^{n-1}\left|y_i - \hat{y}_i\right| \tag{5.12}$$

Here, $\hat{y}$ is the estimated or predicted target output and y is the actual target output, and n is the number of samples used for regression analysis.

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Validation of Regression Methods

Mean Squared Error (MSE)

It is the sum of square of residuals. This value is always positive and closer to 0. This is given mathematically as:

$$\frac{1}{n}\sum_{i=0}^{n-1}(y_i - \hat{y}_i)^2 \qquad (5.13)$$

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Validation of Regression Methods

Root Mean Square Error (RMSE)

The square root of the MSE is called RMSE. This is given as:

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n}\sum_{i=0}^{n-1}(y_i - \hat{y}_i)^2} \tag{5.14}$$

Relative MSE

Relative MSE is the ratio of the prediction ability of the $\hat{y}$ to the average of the trivial population. The value of zero indicates that the model is perfect and its value ranges between 0 and 1. If the value is more than 1, then the created model is not a good one. This is given as follows:

$$\text{RelMSE} = \frac{\sum_{i=0}^{n-1}(y_i - \hat{y}_i)^2}{\sum_{i=0}^{n-1}(y_i - \overline{y}_i)^2} \tag{5.15}$$

Coefficient of Variation

Coefficient of variation is unit less and is given as:

$$\text{CV} = \frac{\text{RMSE}}{\overline{y}} \tag{5.16}$$

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Example

Example 5.3: Consider the following training set Table 5.4 for predicting the sales of the items.

Table 5.4: Training Item Table

Items x_i	Actual Sales (In Thousands) y_i
I_1	80
I_2	90
I_3	100
I_4	110
I_5	120

Consider two fresh items I_6 and I_7, whose actual values are 80 and 75, respectively. A regression model predicts the values of the items I_6 and I_7 as 75 and 85, respectively. Find MAE, MSE, RMSE, RelMSE and CV.

Solution: The test items' actual and prediction is given in Table 5.5 as:

Table 5.5: Test Item Table

Test Items	Actual Value y_i	Predicted Value $\bar{y}_i$
I_6	80	75
I_7	75	85

Mean Absolute Error (MAE) using Eq. (5.12) is given as:

$$MAE = \frac{1}{2} \times |80 - 75| + |75 - 85| = \frac{15}{2} = 7.5$$

Mean Squared Error (MSE) using Eq. (5.13) is given as:

$$MSE = \frac{1}{2} \times |80 - 75|^2 + |75 - 85|^2 = \frac{125}{2} = 62.5$$

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Example

Root Mean Square error using Eq. (5.14) is given as:

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{62.5} = 7.91$$

For finding RelMSE and CV, the training table should be used to find the average of y.

The average of y is $\frac{80 + 90 + 100 + 110 + 120}{5} = \frac{500}{5} = 100.$

RelMSE using Eq. (5.15) can be computed as:

$$\text{RelMSE} = \frac{(80 - 75)^2 + (75 - 85)^2}{(80 - 100)^2 + (75 - 100)^2} = \frac{125}{1025} = 0.1219$$

CV can be computed using Eq. (5.16) as $\frac{\sqrt{62.5}}{100} = 0.08.$

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Validation of Regression Methods

COEFFICIENT OF DETERMINATION

$$r^2 = \frac{\text{Explained variation}}{\text{Total variation}}$$

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Multiple Regression Model

Multiple regression model involves multiple predictors or independent variables and one dependent variable. This is an extension of the linear regression problem. The basic assumptions of multiple linear regression are that the independent variables are not highly correlated and hence multicollinearity problem does not exist. Also, it is assumed that the residuals are normally distributed.

For example, the multiple regression of two variables x_1 and x_2 is given as follows:

$$y = f(x_1, x_2)$$
$$= a_0 + a_1 x_1 + a_2 x_2 \quad (5.21)$$

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Example

Example 5.5: Apply multiple regression for the values given in Table 5.7 where weekly sales along with sales for products x_1 and x_2 are provided. Use matrix approach for finding multiple regression.

Table 5.7: Sample Data

x_1 (Product One Sales)	x_2 (Product Two Sales)	y Output Weekly Sales (in Thousands)
1	4	1
2	5	6
3	8	8
4	2	12

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Example

Solution: Here, the matrices for Y and X are given as follows:

$$X = \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} \text{ and } Y = \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix}.$$

The coefficient of the multiple regression equation is given as $a = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$.

The regression coefficient for multiple regression is calculated the same way as linear regression:

$$\hat{a} = ((X^T X)^{-1} X^T) Y \tag{5.23}$$

Using Eq. (5.23), and substituting the values (Similar to Problem 5.2), one gets $\hat{a}$ as:

$$= \left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} \right)^{-1} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix}$$

$$= \begin{pmatrix} -1.69 \\ 3.48 \\ -0.05 \end{pmatrix}$$

Here, the coefficients are $a_0 = -1.69$, $a_1 = 3.48$ and $a_2 = -0.05$. Hence, the constructed model is:

$y = -1.69 + 3.48x_1 - 0.05x_2$

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Polynomial Regression

The first method is called transformation. The trick is to convert non-linear data to linear data that can be handled using the linear regression method. Let us consider an exponential function $y = ae^{bx}$. The transformation can be done by applying log function to both sides to get:

$$\ln y = bx + \ln a \tag{5.24}$$

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Polynomial Regression

$$y = a_0 + a_1 x + a_2 x^2$$

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma(x_i, y_i) \\ \Sigma(x_i^2, y_i) \end{bmatrix} \quad (5.28)$$

This is of the form $Xa = B$. One can solve this equation for a as:

$$a = X^{-1}B \quad (5.29)$$

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Example

Table 5.8: Sample Data

x	y
1	1
2	4
3	9
4	15

Solution: For applying polynomial regression, computation is done as shown in Table 5.9. Here, the order is 2 and the sample i ranges from 1 to 4.

Table 5.9: Computation Table

x_i	y_i	$x_i y_i$	x_i^2	$x_i^2 y$	x_i^3	x_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
$\Sigma x_i = 10$	$\Sigma y_i = 29$	$\Sigma x_i y_i = 96$	$\Sigma x_i^2 = 30$	$\Sigma x_i^2 y_i = 338$	$\Sigma x_i^3 = 100$	$\Sigma x_i^4 = 354$

It can be noted that, $N = 4$, $\Sigma y_i = 29$, $\Sigma x_i y_i = 96$, $\Sigma x_i^2 y_i = 338$. When the order is 2, the matrix using Eq. (5.28) is given as follows:

$$\begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix}$$

Therefore, using Eq. (5.29), one can get coefficients as:

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{pmatrix} -0.75 \\ 0.95 \\ 0.75 \end{pmatrix}.$$

This leads to the regression equation using Eq. (5.26) as:

$$y = -0.75 + 0.95\,x + 0.75\,x^2$$

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Logistic Regression

1. Is the mail spam or not spam? The answer is yes or no. Thus, categorical dependant variable is a binary response of yes or no.
2. If the student should be admitted or not is based on entrance examination marks. Here, categorical variable response is admitted or not.
3. The student being pass or fail is based on marks secured.

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Logistic Regression

Thus, logistic regression is used as a binary classifier and works by predicting the probability of the categorical variable. In general, it takes one or more features x and predicts the response y. If the probability is predicted via linear regression, it is given as:

$$p(x) = a_0 + a_1 x$$

Hence, logistic regression tries to model the probability of the particular response variable. In email classification problem, say normal email or spam, if the probability of the response variable is 0.7, then there is a 70% possibility of a normal mail.

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Logistic Regression

$$odd = \frac{probability\ \text{of an event}}{probability\ \text{of an non}-\text{event}} = \frac{p}{1-p} \tag{5.31}$$

Log-odds can be taken for the odds, resulting in:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = a_0 + a_1 x \tag{5.32}$$

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Logistic Regression

$$p(x) = \frac{\exp(a_0 + a_1 x)}{1 + \exp(a_0 + a_1 x)} \tag{5.33}$$

This is the same sigmoidal function. It always gives the value in the range 0–1. Dividing the numerator and denominator by the numerator, one gets:

$$p(x) = \frac{1}{1 + \exp(-a_0 - a_1 x)} \tag{5.34}$$

One can rearrange this by taking the minus sign outside to get the following logistic function:

$$p(x) = \frac{1}{1 + \exp(-(a_0 + a_1 x))} \tag{5.35}$$

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Logistic Regression

Here, x is the explanatory or predictor variable, e is the Euler number, and a_0, a_1 are the regression coefficients. The coefficients a_0, a_1 can be learned and the predictor predicts $p(x)$ directly using the threshold function as:

$$y = \begin{cases} 1 & \text{if } p(x) \geq 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (5.36)$$

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Example

Example 5.7: Let us assume a binomial logistic regression problem where the classes are pass and fail. The student dataset has entrance mark based on the historic data of those who are selected or not selected. Based on the logistic regression, the values of the learnt parameters are $a_0 = 1$ and $a_1 = 8$. Assuming marks of $x = 60$, compute the resultant class.

Solution: The values of regression coefficients are $a_0 = 1$ and $a_1 = 8$, and given that $x = 60$.

Based on the regression coefficients, z can be computed as:

$$z = a_0 + a_1 x$$
$$= 1 + 8 \times 60 = 481$$

One can fit this in a sigmoidal function using Eq. (5.30) to get the probability as:

$$y = \frac{1}{1 + \exp(-481)} = \frac{1}{2.271} = 0.44$$

If we assume the threshold value as 0.5, then it is observed that $0.44 < 0.5$, therefore, the candidate with marks 60 is not selected.

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Regularization Methods

There are two issues that need to be considered before discussing regularization methods. One is bias and another is variance.

1. Bias – If the selected model does not fit the dataset well, then this error is called bias.
2. Variance – If the model works nicely for the trained data but is not representative for the entire universe of the possible data, it is called variance.

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Ridge

Sum of squared residuals + $\lambda \times slope^2$

That is, $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \times slope^2$ (5.38)

Here, $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ is the sum of the squared residuals and $\lambda \times slope^2$ is the penalty term added

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Procedure

1. Choose a possible set of values of the penalty
2. Exclude the Nth observed data and compute the penalty

3. Compute the penalty out of samples that are excluded
4. Compute the mean square error and pick the penalty that minimizes the MSE

The matrix formulation of ridge regression is given as follows:

$$\hat{a} = (X'X + kI)^{-1} X'Y \tag{5.39}$$

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Lasso

LASSO regression is better than ridge regression. LASSO stands for 'Least Absolute Shrinkage and Selection Operator'. This uses L_1 regularization.

This adds a penalty factor that equals the square of the magnitude of all coefficients. This penalty results in a simple model given as:

$$\text{Sum of squared residuals} + \lambda \times |slope|$$

Thus, LASSO model is given as:

$$\hat{a} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \sum_{i=0}^{k}|a_i| \tag{5.40}$$

By controlling the factor λ, where the high values of λ force many coefficients to zero, LASSO performs both shrinkage and variable selection.

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Ridge Vs Lasso

Table 5.10: Difference between Ridge and LASSO Regression

Ridge Regression	LASSO Regression
Penalty term is $\lambda \times slope^2$	Penalty term is $\lambda \times \|slope\|$
This method shrinks the regression coefficients of less important variables closer to zero.	This method shrinks the regression coefficients of less important variables to zero giving a compact model.
Not useful for feature selection	Good feature selector by removing all irrelevant variables

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Elastic Net

Elastic Net is a hybrid method of combining both Ridge and LASSO regression methods. The Elastic Net is given as follows:

$$\text{Sum of squared residuals} + \lambda_1 + \underbrace{|v_1| + |v_2| + \cdots + |v_k|}_{Lasso} + \lambda_2 \times \underbrace{|v_1|^2 + |v_2|^2 + \cdots + |v_k|^2}_{Ridge} \quad (5.41)$$

Here, $v_1, v_2, \ldots, v_k$ are dependent variables of the regression method.

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Summary

1. Regression is a supervised learning method that can predict continuous variables.
2. Regression analysis is used to model the relationship between one or more independent variables and a dependent variable whereas in multiple regression problems, the output is a combination of predictor variables.
3. Regression is used for prediction and forecasting. This determines the change in response variables when one exploration variable is varied while keeping all other parameters constant. This is used to determine the relationship each of the exploratory variables exhibits.
4. Scatter plot is a plot of explanatory variable and response variable. It is a 2D graph showing the relationship between two variables. The quality of the regression analysis is determined by the factors of correlation and causation.
5. Linear regression model can be created by fitting a line among the scattered data points. The line is of the form: $y = a_0 + a_1 x$. Here, a_0 is the intercept which represents the bias and a_1 represents the slope of the line.

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6. Multiple regression model is an extension of linear regression model that involves multiple predictors. It is also used to find relationships among variables, important features and construct models.
7. Polynomial regression can handle non-linear relationships among variables by using degrees of polynomial to make non-regression line. If the relationship is not linear, then polynomial regression can be used at different levels of curvilinearity.
8. Logistic regression is suitable for binary classification problem. Here, the output is often a binary variable. For example, consider a question – Is the mail spam or not spam? The answer is yes or no. Thus, logistic regression is used to predict categorical dependant variable.
9. Ridge regression is used to create a parsimonious model by introducing ridge estimator that introduces some bias in the model to reduce variance.
10. LASSO stands for "Least Absolute Shrinkage and Selection Operator". This uses L_1 regularization to remove irrelevant variables.

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